5.5.3. Phase Sequence

It is the term which is used to represent in what sequence the three phase voltage or current attains maximum value. If the phase sequence is said to be R, Y, B then first red phase attains maximum value with a phase difference of 120° each, the yellow phase and blue phase attains their peak value.

5.5.4. Inter Connection of Three Phases

In a three phase alternator, there are three windings or phases. Each phase has two terminals. If a separate load is connected across each winding six conductors are required to transmit power. This will make the system complicated and expensive. In practice, three windings are interconnected to give two methods of connection.

Star (Y) connection

Mesh (Δ) connection.

5.5.5. Star Y Connection

In this method, similar ends of the three phases are joined together to form a common junction (N) supply is taken from other three ends. The common junction (N) is called the star point or netural point. The voltage between any one line and netural is called phase voltage. The current flows through that phase is called phase current. Voltage between any two lines is called line voltage and current through that line is called line current.

In this connection,

Phase current = Line current

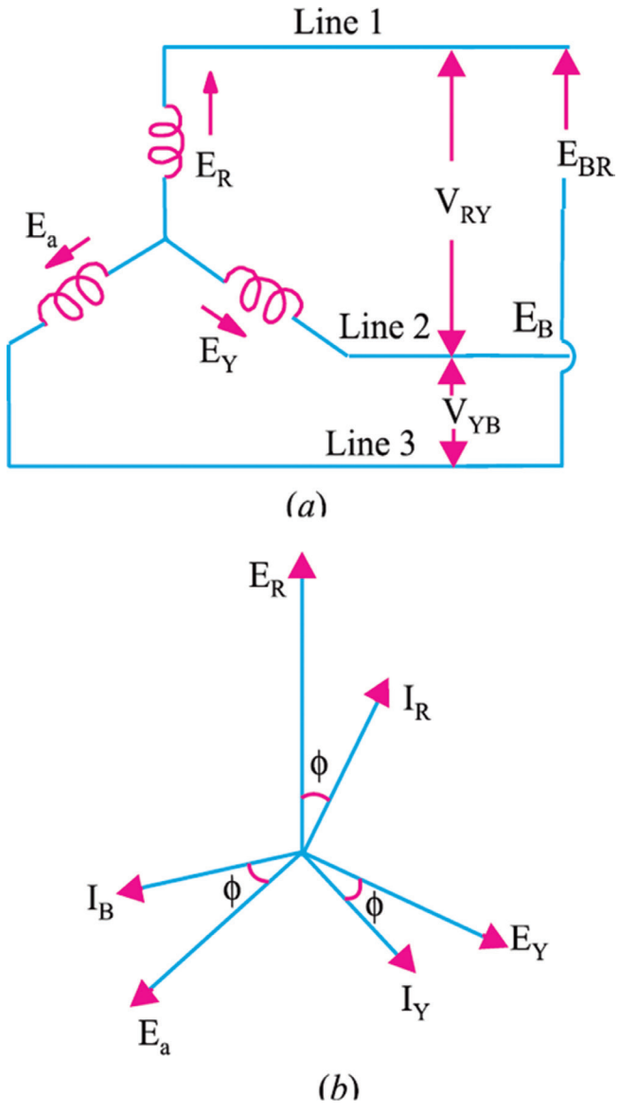


Fig 5.14 Star Connection

$$I_{ph} = I_L$$

$$\text{Phase voltage} = \frac{\text{Line Voltage}}{\sqrt{3}} = \frac{E_L}{\sqrt{3}}$$

If the neutral wire is taken for external connection, then the system is called a three-phase four wire star connected system. If the neutral wire is not taken for external connection, then the system is called a three phase three wire star connected system.

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